

Practical no : 4

// Online C++ compiler to run C++ program online

```
#include <iostream>
```

```
using namespace std;
```

```
void display(int a[])
```

```
{
```

```
    for(int i=0;i<5;i++)
```

```
    {
```

```
        cout<< a[i] <<" ";
```

```
        cout<<endl;
```

```
    }
```

```
}
```

```
int main()
```

```
{
```

```
    int a[5]={252,23,34,110,9};
```

```
    int b[5],c[5],temp;
```

```
    for(int i=0;i<5;i++)
```

```
    {
```

```
        b[i]=a[i];
```

```
        c[i]=a[i];
```

```
    }
```

```
//BUBBLE SORT
```

```
for (int i=0;i<4;i++)
```

```
{
```

```
    for(int j=0;j<4-i;j++){
```

```
        if (a[j]>a[j+1])
```

```
            swap(a[j],a[j+1]);
```

```

    }
}
// selection sort
for (int i=0;i<4;i++)
{
    int min=i;
    for(int j=i+1;j<5;j++)

        if(b[j]<b[min])

            min=j;

            swap(b[i],b[min]);
}
//insertion sort

for(int i=1;i<5;i++)
{
    temp=c[i];
    int j=i-1;

    while(j>=0&& c[j]>temp){
        c[j+1]=c[j];
        j--;
    }

    c[j+1]=temp;

}
cout<<"\n bubble sort=";
```

```
display(a);

cout<<"\n selection sort:";
display(b);

cout<<"\n insertion sort:";
display(c);
return 0;
}
```

Output:

Bubble sort:9

23

34

110

252

selection sort:9

23

34

110

252

insertion sort:9

23

34

110

252